

Section 1 - Free Text Queries

Query

cotton kurta for men

Options

☐ Use proximity-aware ranking ⓘ

☐ Automatically correct spelling ⓘ

Search

Top 10 result(s) — Inc.ltc baseline ranking.

Rank	Doc ID	Title	Category	Cosine score
1	D004	Men's Regular Fit Kurta - Pink	Kurta	0.2755
2	D064	Men's Regular Fit Kurta - White	Kurta	0.2755
3	D024	Men's Regular Fit Kurta - Teal	Kurta	0.2730
4	D084	Men's Regular Fit Kurta - Beige	Kurta	0.2730
5	D044	Men's Regular Fit Kurta - Navy Blue	Kurta	0.2648
6	D034	Women's Printed Straight Kurta - Maroon	Kurta	0.2255
7	D094	Women's Printed Straight Kurta - Teal	Kurta	0.2255
8	D054	Women's Printed Straight Kurta - Mustard	Kurta	0.2235
9	D014	Women's Printed Straight Kurta - Beige	Kurta	0.2218
10	D074	Women's Printed Straight Kurta - Pink	Kurta	0.2218

Query

slim fit jeans denim

Options

☐ Use proximity-aware ranking ⓘ

☐ Automatically correct spelling ⓘ

Search

Top 10 result(s) — Inc.ltc baseline ranking.

Rank	Doc ID	Title	Category	Cosine score
1	D053	Men's Slim Fit Stretch Jeans - Grey	Jeans	0.3510
2	D013	Men's Slim Fit Stretch Jeans - Grey	Jeans	0.3506
3	D073	Men's Slim Fit Stretch Jeans - Grey	Jeans	0.3506
4	D033	Men's Slim Fit Stretch Jeans - Grey	Jeans	0.3446
5	D093	Men's Slim Fit Stretch Jeans - Grey	Jeans	0.3446
6	D023	Men's Regular Fit Denim Jeans - Grey	Jeans	0.2918
7	D083	Men's Regular Fit Denim Jeans - Grey	Jeans	0.2918
8	D043	Men's Regular Fit Denim Jeans - Grey	Jeans	0.2899
9	D003	Men's Regular Fit Denim Jeans - Grey	Jeans	0.2849
10	D063	Men's Regular Fit Denim Jeans - Grey	Jeans	0.2849

ClothingSpree Search

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Ask Gemini

Deploy

Query

women ethnic festive wear

Options

☐ Use proximity-aware ranking

☐ Automatically correct spelling

Search

Top 10 result(s) — Inc.ltc baseline ranking.

Rank	Doc ID	Title	Category	Cosine score
1	D076	Women's Casual Fit Dress - Green	Dress	0.1991
2	D058	Women's Quilted Winter Jacket - Black	Jacket	0.1975
3	D036	Women's Casual Fit Dress - Black	Dress	0.1973
4	D018	Women's Quilted Winter Jacket - Beige	Jacket	0.1961
5	D078	Women's Quilted Winter Jacket - Beige	Jacket	0.1961
6	D010	Women's Fleece Sweatshirt - Black	Sweatshirt	0.1957
7	D070	Women's Fleece Sweatshirt - Black	Sweatshirt	0.1957
8	D028	Women's Casual Puffer Jacket - Black	Jacket	0.1953
9	D088	Women's Casual Puffer Jacket - Black	Jacket	0.1953
10	D056	Women's Casual Fit Dress - Yellow	Dress	0.1950

ClothingSpree Search

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Ask Gemini

Deploy

Query

polo neck t-shirt casual

Options

☐ Use proximity-aware ranking

☐ Automatically correct spelling

Search

Top 10 result(s) — Inc.ltc baseline ranking.

Rank	Doc ID	Title	Category	Cosine score
1	D001	Men's Cotton Crew Neck T-Shirt - Black	T-Shirt	0.2691
2	D041	Men's Cotton Crew Neck T-Shirt - Maroon	T-Shirt	0.2691
3	D061	Men's Cotton Crew Neck T-Shirt - White	T-Shirt	0.2691
4	D081	Men's Cotton Crew Neck T-Shirt - Mustard	T-Shirt	0.2663
5	D021	Men's Cotton Crew Neck T-Shirt - Sky Blue	T-Shirt	0.2605
6	D052	Women's Striped Casual Shirt - Beige	Shirt	0.1046
7	D022	Men's Checked Cotton Shirt - White	Shirt	0.1040
8	D082	Men's Checked Cotton Shirt - Blue	Shirt	0.1040
9	D012	Women's Striped Casual Shirt - Blue	Shirt	0.1038
10	D072	Women's Striped Casual Shirt - Black	Shirt	0.1038

ClothingSpree Search

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Ask Gemini

Deploy

Query

formal shirt office wear

Options

☐ Use proximity-aware ranking

☐ Automatically correct spelling

Search

Top 10 result(s) — Inc.Itc baseline ranking.

Rank	Doc ID	Title	Category	Cosine score
1	D052	Women's Striped Casual Shirt - Beige	Shirt	0.2209
2	D022	Men's Checked Cotton Shirt - White	Shirt	0.2197
3	D082	Men's Checked Cotton Shirt - Blue	Shirt	0.2197
4	D012	Women's Striped Casual Shirt - Blue	Shirt	0.2194
5	D072	Women's Striped Casual Shirt - Black	Shirt	0.2194
6	D011	Men's Oversized Graphic T-Shirt - Mustard	T-Shirt	0.2182
7	D042	Men's Checked Cotton Shirt - Maroon	Shirt	0.2182
8	D071	Men's Oversized Graphic T-Shirt - Black	T-Shirt	0.2182
9	D092	Women's Striped Casual Shirt - White	Shirt	0.2170
10	D002	Men's Checked Cotton Shirt - Black	Shirt	0.2159

ClothingSpree Search

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Ask Gemini

Deploy

Query

winter jacket warm hooded

Options

☐ Use proximity-aware ranking

☐ Automatically correct spelling

Search

Top 10 result(s) — Inc.Itc baseline ranking.

Rank	Doc ID	Title	Category	Cosine score
1	D058	Women's Quilted Winter Jacket - Black	Jacket	0.2709
2	D018	Women's Quilted Winter Jacket - Beige	Jacket	0.2690
3	D078	Women's Quilted Winter Jacket - Beige	Jacket	0.2690
4	D038	Women's Quilted Winter Jacket - Olive	Jacket	0.2660
5	D098	Women's Quilted Winter Jacket - Olive	Jacket	0.2660
6	D028	Women's Casual Puffer Jacket - Black	Jacket	0.2323
7	D088	Women's Casual Puffer Jacket - Black	Jacket	0.2323
8	D048	Women's Casual Puffer Jacket - Beige	Jacket	0.2307
9	D008	Women's Casual Puffer Jacket - Olive	Jacket	0.2282
10	D068	Women's Casual Puffer Jacket - Olive	Jacket	0.2282

ClothingSpree Search

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Deploy

Query

breathable fabric summer

Options

☐ Use proximity-aware ranking

☐ Automatically correct spelling

Search

Top 10 result(s) — Inc.Itc baseline ranking.

Rank	Doc ID	Title	Category	Cosine score
1	D014	Women's Printed Straight Kurta - Beige	Kurta	0.2073
2	D074	Women's Printed Straight Kurta - Pink	Kurta	0.2073
3	D044	Men's Regular Fit Kurta - Navy Blue	Kurta	0.2048
4	D004	Men's Regular Fit Kurta - Pink	Kurta	0.1958
5	D064	Men's Regular Fit Kurta - White	Kurta	0.1958
6	D024	Men's Regular Fit Kurta - Teal	Kurta	0.1940
7	D084	Men's Regular Fit Kurta - Beige	Kurta	0.1940
8	D034	Women's Printed Straight Kurta - Maroon	Kurta	0.1936
9	D094	Women's Printed Straight Kurta - Teal	Kurta	0.1936
10	D011	Men's Oversized Graphic T-Shirt - Mustard	T-Shirt	0.1933

ClothingSpree Search

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Deploy

Query

high waist palazzo trousers

Options

☐ Use proximity-aware ranking

☐ Automatically correct spelling

Search

Top 10 result(s) — Inc.Itc baseline ranking.

Rank	Doc ID	Title	Category	Cosine score
1	D029	Women's High Waist Stretch Leggings - Grey	Leggings	0.3057
2	D049	Women's High Waist Stretch Leggings - Black	Leggings	0.3057
3	D089	Women's High Waist Stretch Leggings - Grey	Leggings	0.3057
4	D009	Women's High Waist Stretch Leggings - Olive Green	Leggings	0.2958
5	D069	Women's High Waist Stretch Leggings - Olive Green	Leggings	0.2958
6	D019	Women's Printed Active Leggings - Black	Leggings	0.2073
7	D059	Women's Printed Active Leggings - Grey	Leggings	0.2073
8	D079	Women's Printed Active Leggings - Black	Leggings	0.2073
9	D039	Women's Printed Active Leggings - Olive Green	Leggings	0.2006
10	D099	Women's Printed Active Leggings - Olive Green	Leggings	0.2006

ClothingSpree Search x + Ask Gemini

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Deploy

Query

embroidered saree silk

Options

☐ Use proximity-aware ranking ⓘ

☐ Automatically correct spelling ⓘ

Search

Top 10 result(s) — Inc.Itc baseline ranking.

Rank	Doc ID	Title	Category	Cosine score
1	D035	Women's Cotton Handloom Saree - Yellow	Saree	0.2197
2	D055	Women's Cotton Handloom Saree - Pink	Saree	0.2197
3	D095	Women's Cotton Handloom Saree - Maroon	Saree	0.2197
4	D005	Women's Printed Daily Wear Saree - Blue	Saree	0.2188
5	D025	Women's Printed Daily Wear Saree - Maroon	Saree	0.2188
6	D065	Women's Printed Daily Wear Saree - Green	Saree	0.2188
7	D085	Women's Printed Daily Wear Saree - Red	Saree	0.2188
8	D015	Women's Cotton Handloom Saree - Red	Saree	0.2173
9	D075	Women's Cotton Handloom Saree - Blue	Saree	0.2173
10	D045	Women's Printed Daily Wear Saree - Purple	Saree	0.2164

ClothingSpree Search x + Ask Gemini

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Deploy

Query

zip closure pocket cargo

Options

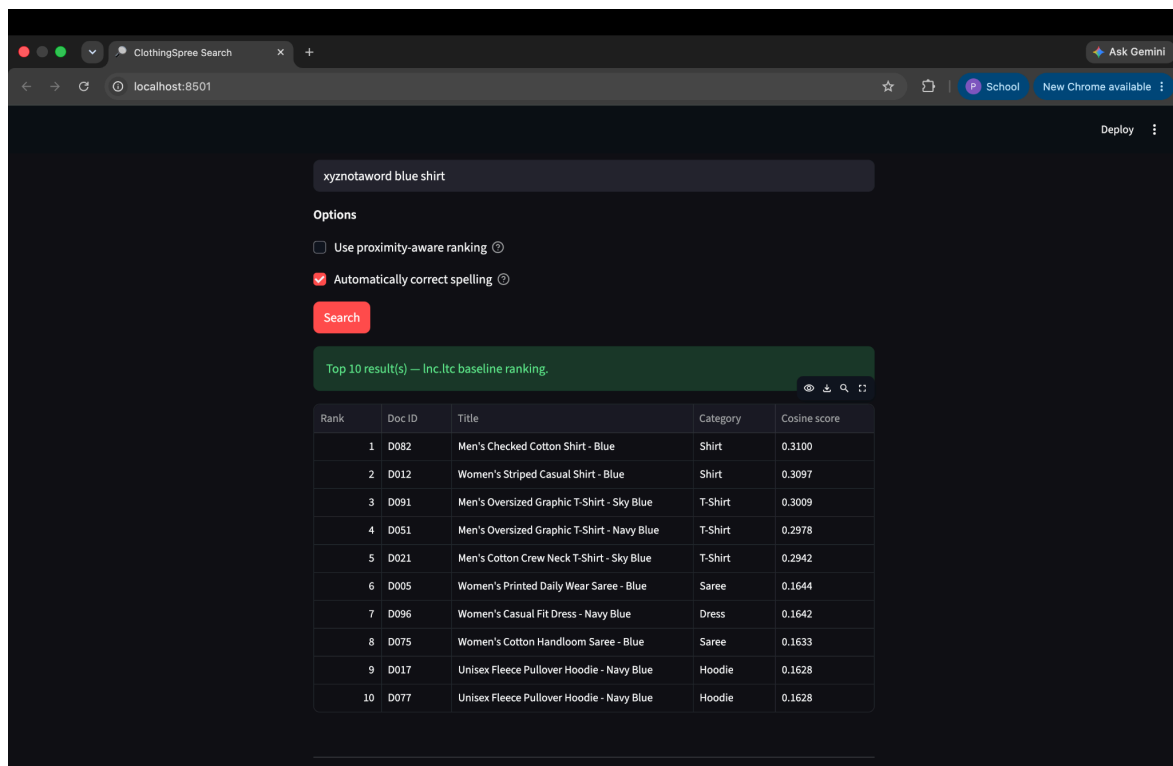
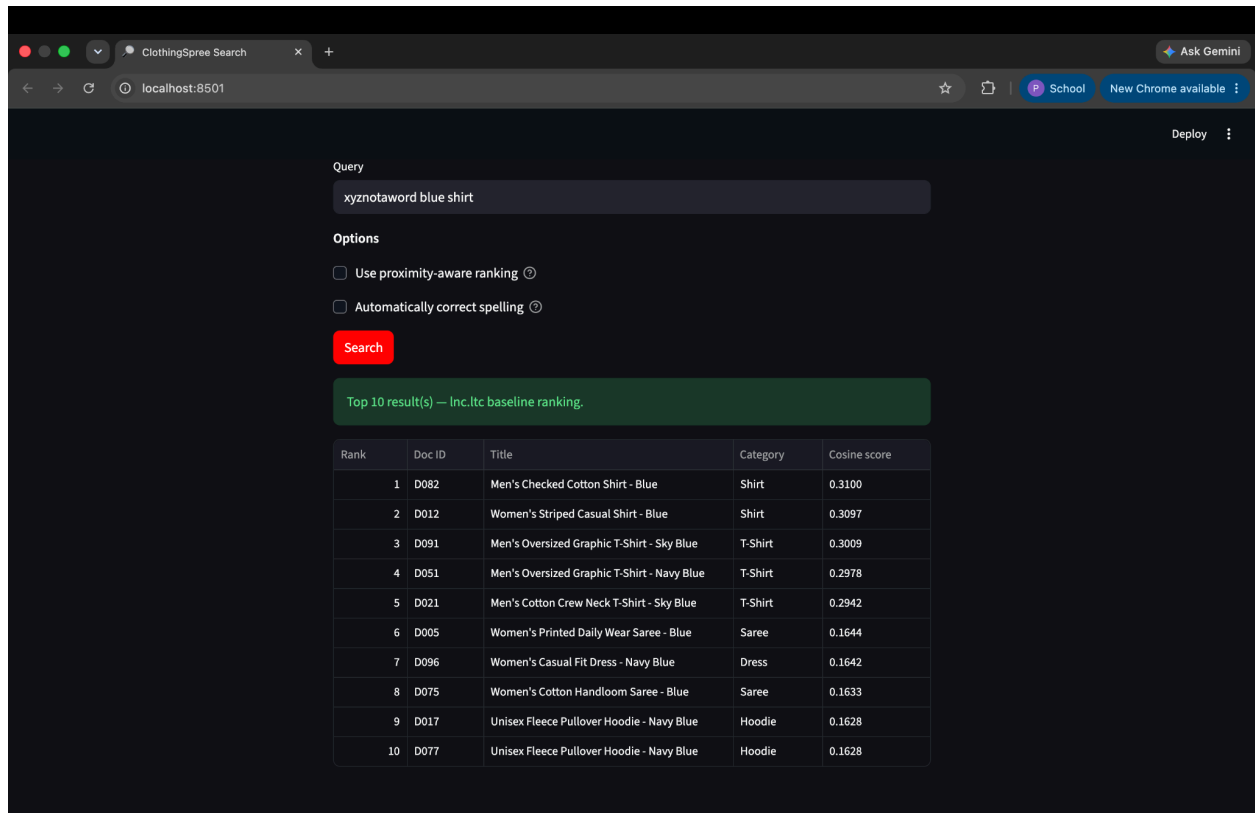
☐ Use proximity-aware ranking ⓘ

☐ Automatically correct spelling ⓘ

Search

Top 10 result(s) — Inc.Itc baseline ranking.

Rank	Doc ID	Title	Category	Cosine score
1	D037	Unisex Fleece Pullover Hoodie - Black	Hoodie	0.0867
2	D097	Unisex Fleece Pullover Hoodie - Black	Hoodie	0.0867
3	D052	Women's Striped Casual Shirt - Beige	Shirt	0.0864
4	D022	Men's Checked Cotton Shirt - White	Shirt	0.0859
5	D082	Men's Checked Cotton Shirt - Blue	Shirt	0.0859
6	D012	Women's Striped Casual Shirt - Blue	Shirt	0.0858
7	D057	Unisex Fleece Pullover Hoodie - Wine	Hoodie	0.0858
8	D072	Women's Striped Casual Shirt - Black	Shirt	0.0858
9	D007	Women's Oversized Fleece Hoodie - Black	Hoodie	0.0853
10	D042	Men's Checked Cotton Shirt - Maroon	Shirt	0.0853



SECTION 2 — Exact Phrase Queries

ClothingSpree Search

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Deploy

☒ Exact phrase search

☐ Proximity search (WITHIN/k)

Exact phrase search — all words must appear in the given order at consecutive positions. Mere co-occurrence of the words in a document is not enough.

Phrase

cotton shirt

Search phrase

Found the exact phrase in 10 document(s). Each position group is [start, start+1, ...] — strictly consecutive.

Doc ID	Title	Category	Matching positions
D001	Men's Cotton Crew Neck T-Shirt - Black	T-Shirt	[14, 15]
D002	Men's Checked Cotton Shirt - Black	Shirt	[2, 3] • [7, 8]
D021	Men's Cotton Crew Neck T-Shirt - Sky Blue	T-Shirt	[16, 17]
D022	Men's Checked Cotton Shirt - White	Shirt	[2, 3] • [7, 8]
D041	Men's Cotton Crew Neck T-Shirt - Maroon	T-Shirt	[14, 15]
D042	Men's Checked Cotton Shirt - Maroon	Shirt	[2, 3] • [7, 8]
D061	Men's Cotton Crew Neck T-Shirt - White	T-Shirt	[14, 15]
D062	Men's Checked Cotton Shirt - Olive	Shirt	[2, 3] • [7, 8]
D081	Men's Cotton Crew Neck T-Shirt - Mustard	T-Shirt	[14, 15]
D082	Men's Checked Cotton Shirt - Blue	Shirt	[2, 3] • [7, 8]

ClothingSpree Search

Ask Gemini

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Deploy

☒ Exact phrase search

☐ Proximity search (WITHIN/k)

Exact phrase search — all words must appear in the given order at consecutive positions. Mere co-occurrence of the words in a document is not enough.

Phrase

stretch denim

Search phrase

Found the exact phrase in 7 document(s). Each position group is [start, start+1, ...] — strictly consecutive.

Doc ID	Title	Category	Matching positions
D003	Men's Regular Fit Denim Jeans - Grey	Jeans	[14, 15]
D013	Men's Slim Fit Stretch Jeans - Grey	Jeans	[13, 14]
D033	Men's Slim Fit Stretch Jeans - Grey	Jeans	[14, 15]
D043	Men's Regular Fit Denim Jeans - Grey	Jeans	[13, 14]
D063	Men's Regular Fit Denim Jeans - Grey	Jeans	[14, 15]
D073	Men's Slim Fit Stretch Jeans - Grey	Jeans	[13, 14]
D093	Men's Slim Fit Stretch Jeans - Grey	Jeans	[14, 15]

ClothingSpree Search

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Deploy

Search type

Exact phrase search

Proximity search (WITHIN/k)

Exact phrase search

all words must appear in the given order at consecutive positions. Mere co-occurrence of the words in a document is not enough.

Phrase

festive wear

Search phrase

Found the exact phrase in 10 document(s). Each position group is [start, start+1, ...] – strictly consecutive.

Doc ID	Title	Category	Matching positions
D005	Women's Printed Daily Wear Saree - Blue	Saree	[20, 21]
D015	Women's Cotton Handloom Saree - Red	Saree	[18, 19]
D025	Women's Printed Daily Wear Saree - Maroon	Saree	[20, 21]
D035	Women's Cotton Handloom Saree - Yellow	Saree	[18, 19]
D045	Women's Printed Daily Wear Saree - Purple	Saree	[20, 21]
D055	Women's Cotton Handloom Saree - Pink	Saree	[18, 19]
D065	Women's Printed Daily Wear Saree - Green	Saree	[20, 21]
D075	Women's Cotton Handloom Saree - Blue	Saree	[18, 19]
D085	Women's Printed Daily Wear Saree - Red	Saree	[20, 21]
D095	Women's Cotton Handloom Saree - Maroon	Saree	[18, 19]

ClothingSpree Search

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Deploy

Search type

Exact phrase search

Proximity search (WITHIN/k)

Exact phrase search

all words must appear in the given order at consecutive positions. Mere co-occurrence of the words in a document is not enough.

Phrase

winter wear

Search phrase

Found the exact phrase in 20 document(s). Each position group is [start, start+1, ...] – strictly consecutive.

Doc ID	Title	Category	Matching positions
D008	Women's Casual Puffer Jacket - Olive	Jacket	[20, 21]
D010	Women's Fleece Sweatshirt - Black	Sweatshirt	[17, 18]
D018	Women's Quilted Winter Jacket - Beige	Jacket	[19, 20]
D020	Men's Regular Fit Sweatshirt - Navy Blue	Sweatshirt	[21, 22]
D028	Women's Casual Puffer Jacket - Black	Jacket	[20, 21]
D030	Women's Fleece Sweatshirt - Maroon	Sweatshirt	[17, 18]
D038	Women's Quilted Winter Jacket - Olive	Jacket	[19, 20]
D040	Men's Regular Fit Sweatshirt - Black	Sweatshirt	[19, 20]
D048	Women's Casual Puffer Jacket - Beige	Jacket	[20, 21]
D050	Women's Fleece Sweatshirt - Navy Blue	Sweatshirt	[19, 20]

ClothingSpree Search

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Deploy

Search type

Exact phrase search

Proximity search (WITHIN/k)

Exact phrase search

all words must appear in the given order at consecutive positions. Mere co-occurrence of the words in a document is not enough.

Phrase

regular fit

Search phrase

Found the exact phrase in 15 document(s). Each position group is [start, start+1, ...] — strictly consecutive.

Doc ID	Title	Category	Matching positions
D003	Men's Regular Fit Denim Jeans - Grey	Jeans	[1, 2] • [7, 8]
D004	Men's Regular Fit Kurta - Pink	Kurta	[1, 2] • [6, 7]
D020	Men's Regular Fit Sweatshirt - Navy Blue	Sweatshirt	[1, 2] • [7, 8]
D023	Men's Regular Fit Denim Jeans - Grey	Jeans	[1, 2] • [7, 8]
D024	Men's Regular Fit Kurta - Teal	Kurta	[1, 2] • [6, 7]
D040	Men's Regular Fit Sweatshirt - Black	Sweatshirt	[1, 2] • [6, 7]
D043	Men's Regular Fit Denim Jeans - Grey	Jeans	[1, 2] • [7, 8]
D044	Men's Regular Fit Kurta - Navy Blue	Kurta	[1, 2] • [7, 8]
D060	Men's Regular Fit Sweatshirt - Maroon	Sweatshirt	[1, 2] • [6, 7]
D063	Men's Regular Fit Denim Jeans - Grey	Jeans	[1, 2] • [7, 8]

ClothingSpree Search

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SchoolNew Chrome available

Deploy

Search type

☒ Exact phrase search ☐ Proximity search (WITHIN/k)

Exact phrase search — all words must appear in the given order at consecutive positions. Mere co-occurrence of the words in a document is not enough.

Phrase

breathable fabric

Search phrase

Found the exact phrase in 20 document(s). Each position group is [start, start+1, ...] — strictly consecutive.

Doc ID	Title	Category	Matching positions
D001	Men's Cotton Crew Neck T-Shirt - Black	T-Shirt	[21, 22]
D004	Men's Regular Fit Kurta - Pink	Kurta	[19, 20]
D011	Men's Oversized Graphic T-Shirt - Mustard	T-Shirt	[19, 20]
D014	Women's Printed Straight Kurta - Beige	Kurta	[19, 20]
D021	Men's Cotton Crew Neck T-Shirt - Sky Blue	T-Shirt	[23, 24]
D024	Men's Regular Fit Kurta - Teal	Kurta	[19, 20]
D031	Men's Oversized Graphic T-Shirt - Olive Green	T-Shirt	[21, 22]
D034	Women's Printed Straight Kurta - Maroon	Kurta	[19, 20]
D041	Men's Cotton Crew Neck T-Shirt - Maroon	T-Shirt	[21, 22]
D044	Men's Regular Fit Kurta - Navy Blue	Kurta	[21, 22]

ClothingSpree Search

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SchoolNew Chrome available

Deploy

Phrase and proximity search

These searches use the **positional index**: every token's position in the processed document is stored so we can check whether terms are adjacent (phrase) or within k positions of each other (proximity). Matching positions are shown as evidence.

Positions are zero-based indices into the processed token stream (after stopword removal and stemming).

Search type

☒ Exact phrase search ☐ Proximity search (WITHIN/k)

Exact phrase search — all words must appear in the given order at consecutive positions. Mere co-occurrence of the words in a document is not enough.

Phrase

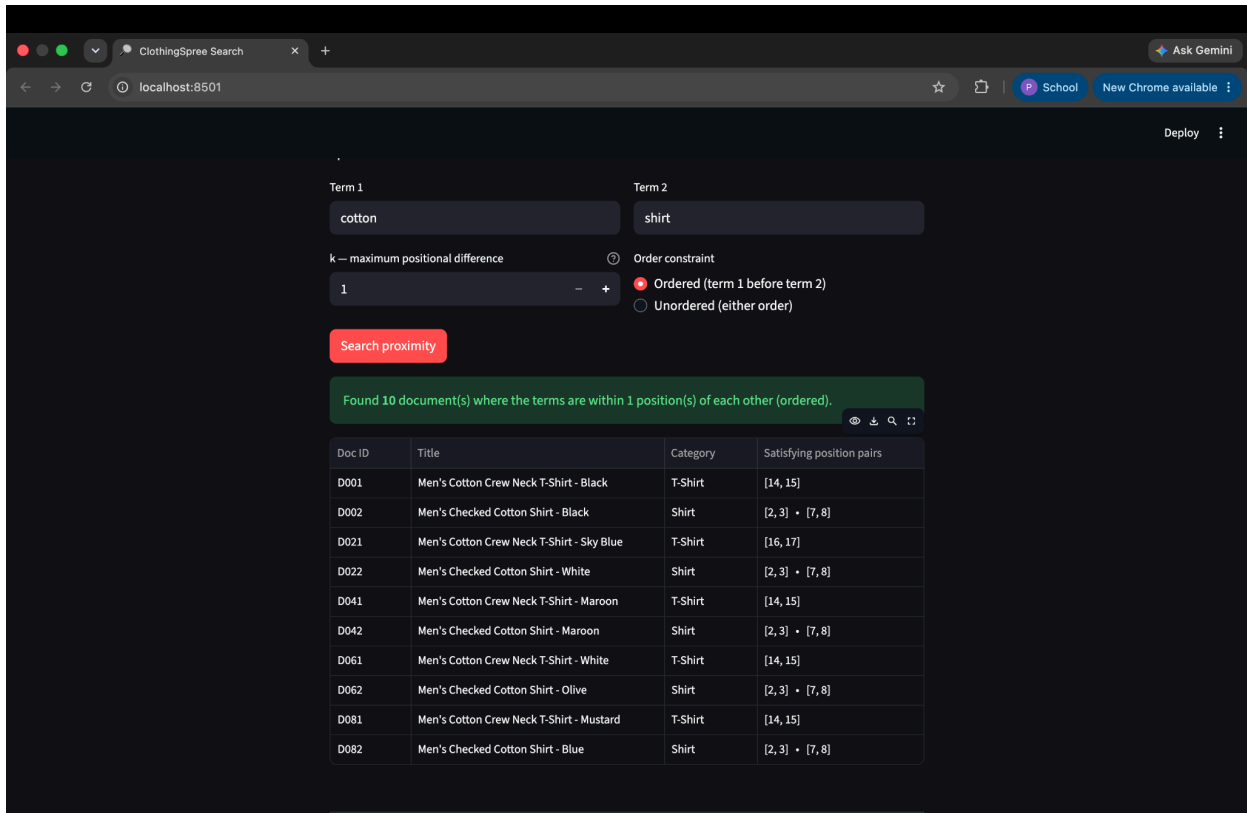
zip closure

Search phrase

No documents contain this exact phrase. A term may be unknown to the corpus vocabulary, or the words never occur adjacently in any document.

> How does this search work?

SECTION 3 — Proximity Queries



Term 1: cotton

Term 2: shirt

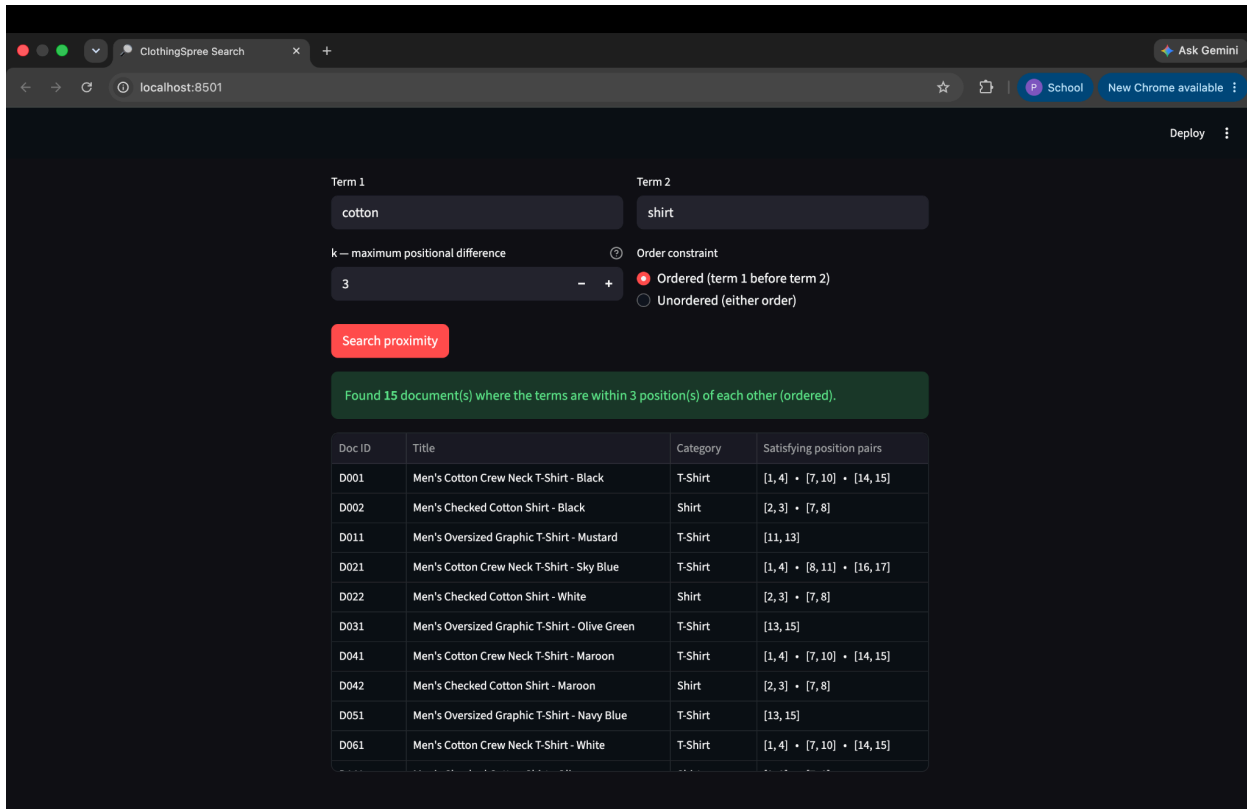
k — maximum positional difference: 1

Order constraint: ☒ Ordered (term 1 before term 2) ☐ Unordered (either order)

Search proximity

Found 10 document(s) where the terms are within 1 position(s) of each other (ordered).

Doc ID	Title	Category	Satisfying position pairs
D001	Men's Cotton Crew Neck T-Shirt - Black	T-Shirt	[14, 15]
D002	Men's Checked Cotton Shirt - Black	Shirt	[2, 3] • [7, 8]
D021	Men's Cotton Crew Neck T-Shirt - Sky Blue	T-Shirt	[16, 17]
D022	Men's Checked Cotton Shirt - White	Shirt	[2, 3] • [7, 8]
D041	Men's Cotton Crew Neck T-Shirt - Maroon	T-Shirt	[14, 15]
D042	Men's Checked Cotton Shirt - Maroon	Shirt	[2, 3] • [7, 8]
D061	Men's Cotton Crew Neck T-Shirt - White	T-Shirt	[14, 15]
D062	Men's Checked Cotton Shirt - Olive	Shirt	[2, 3] • [7, 8]
D081	Men's Cotton Crew Neck T-Shirt - Mustard	T-Shirt	[14, 15]
D082	Men's Checked Cotton Shirt - Blue	Shirt	[2, 3] • [7, 8]



Term 1: cotton

Term 2: shirt

k — maximum positional difference: 3

Order constraint: ☒ Ordered (term 1 before term 2) ☐ Unordered (either order)

Search proximity

Found 15 document(s) where the terms are within 3 position(s) of each other (ordered).

Doc ID	Title	Category	Satisfying position pairs
D001	Men's Cotton Crew Neck T-Shirt - Black	T-Shirt	[1, 4] • [7, 10] • [14, 15]
D002	Men's Checked Cotton Shirt - Black	Shirt	[2, 3] • [7, 8]
D011	Men's Oversized Graphic T-Shirt - Mustard	T-Shirt	[11, 13]
D021	Men's Cotton Crew Neck T-Shirt - Sky Blue	T-Shirt	[1, 4] • [8, 11] • [16, 17]
D022	Men's Checked Cotton Shirt - White	Shirt	[2, 3] • [7, 8]
D031	Men's Oversized Graphic T-Shirt - Olive Green	T-Shirt	[13, 15]
D041	Men's Cotton Crew Neck T-Shirt - Maroon	T-Shirt	[1, 4] • [7, 10] • [14, 15]
D042	Men's Checked Cotton Shirt - Maroon	Shirt	[2, 3] • [7, 8]
D051	Men's Oversized Graphic T-Shirt - Navy Blue	T-Shirt	[13, 15]
D061	Men's Cotton Crew Neck T-Shirt - White	T-Shirt	[1, 4] • [7, 10] • [14, 15]

ClothingSpree Search

Ask Gemini

localhost:8501

Deploy

Term 1

Term 2

cotton

shirt

k — maximum positional difference

5

Order constraint

Ordered (term 1 before term 2)

Unordered (either order)

Search proximity

Found 15 document(s) where the terms are within 5 position(s) of each other (ordered).

Doc ID	Title	Category	Satisfying position pairs
D001	Men's Cotton Crew Neck T-Shirt - Black	T-Shirt	[1, 4] • [7, 10] • [14, 15]
D002	Men's Checked Cotton Shirt - Black	Shirt	[2, 3] • [7, 8]
D011	Men's Oversized Graphic T-Shirt - Mustard	T-Shirt	[11, 13]
D021	Men's Cotton Crew Neck T-Shirt - Sky Blue	T-Shirt	[1, 4] • [8, 11] • [16, 17]
D022	Men's Checked Cotton Shirt - White	Shirt	[2, 3] • [7, 8]
D031	Men's Oversized Graphic T-Shirt - Olive Green	T-Shirt	[13, 15]
D041	Men's Cotton Crew Neck T-Shirt - Maroon	T-Shirt	[1, 4] • [7, 10] • [14, 15]
D042	Men's Checked Cotton Shirt - Maroon	Shirt	[2, 3] • [7, 8]
D051	Men's Oversized Graphic T-Shirt - Navy Blue	T-Shirt	[13, 15]
D061	Men's Cotton Crew Neck T-Shirt - White	T-Shirt	[1, 4] • [7, 10] • [14, 15]

ClothingSpree Search

Ask Gemini

localhost:8501

Deploy

Term 1

Term 2

stretch denim

denim

k — maximum positional difference

4

Order constraint

Ordered (term 1 before term 2)

Unordered (either order)

Search proximity

Found 7 document(s) where the terms are within 4 position(s) of each other (ordered).

Doc ID	Title	Category	Satisfying position pairs
D003	Men's Regular Fit Denim Jeans - Grey	Jeans	[14, 15]
D013	Men's Slim Fit Stretch Jeans - Grey	Jeans	[13, 14]
D033	Men's Slim Fit Stretch Jeans - Grey	Jeans	[14, 15]
D043	Men's Regular Fit Denim Jeans - Grey	Jeans	[13, 14]
D063	Men's Regular Fit Denim Jeans - Grey	Jeans	[14, 15]
D073	Men's Slim Fit Stretch Jeans - Grey	Jeans	[13, 14]
D093	Men's Slim Fit Stretch Jeans - Grey	Jeans	[14, 15]

> How does this search work?

ClothingSpree Search

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Deploy

Term 1

winter

Term 2

wear

k — maximum positional difference

3

Order constraint

☒ Ordered (term 1 before term 2)

☐ Unordered (either order)

Search proximity

Found 20 document(s) where the terms are within 3 position(s) of each other (ordered).

Doc ID	Title	Category	Satisfying position pairs
D008	Women's Casual Puffer Jacket - Olive	Jacket	[20, 21]
D010	Women's Fleece Sweatshirt - Black	Sweatshirt	[17, 18]
D018	Women's Quilted Winter Jacket - Beige	Jacket	[19, 20]
D020	Men's Regular Fit Sweatshirt - Navy Blue	Sweatshirt	[21, 22]
D028	Women's Casual Puffer Jacket - Black	Jacket	[20, 21]
D030	Women's Fleece Sweatshirt - Maroon	Sweatshirt	[17, 18]
D038	Women's Quilted Winter Jacket - Olive	Jacket	[19, 20]
D040	Men's Regular Fit Sweatshirt - Black	Sweatshirt	[19, 20]
D048	Women's Casual Puffer Jacket - Beige	Jacket	[20, 21]
D050	Women's Fleece Sweatshirt - Navy Blue	Sweatshirt	[19, 20]

ClothingSpree Search

Ask Gemini

localhost:8501

Deploy

Phrase and proximity search

These searches use the **positional index**: every token's position in the processed document is stored so we can check whether terms are adjacent (phrase) or within k positions of each other (proximity). Matching positions are shown as evidence.

Positions are zero-based indices into the processed token stream (after stopword removal and stemming).

Search type

☐ Exact phrase search ☒ Proximity search (WITHIN/ k)

Proximity search (WITHIN/ k) — the two terms must occur within k positions of each other. k is the *maximum positional difference*, not the count of words in between: $k = 1$ means adjacent, $k = 3$ allows up to two tokens between them.

Term 1

festive

Term 2

kurta

k — maximum positional difference

4

Order constraint

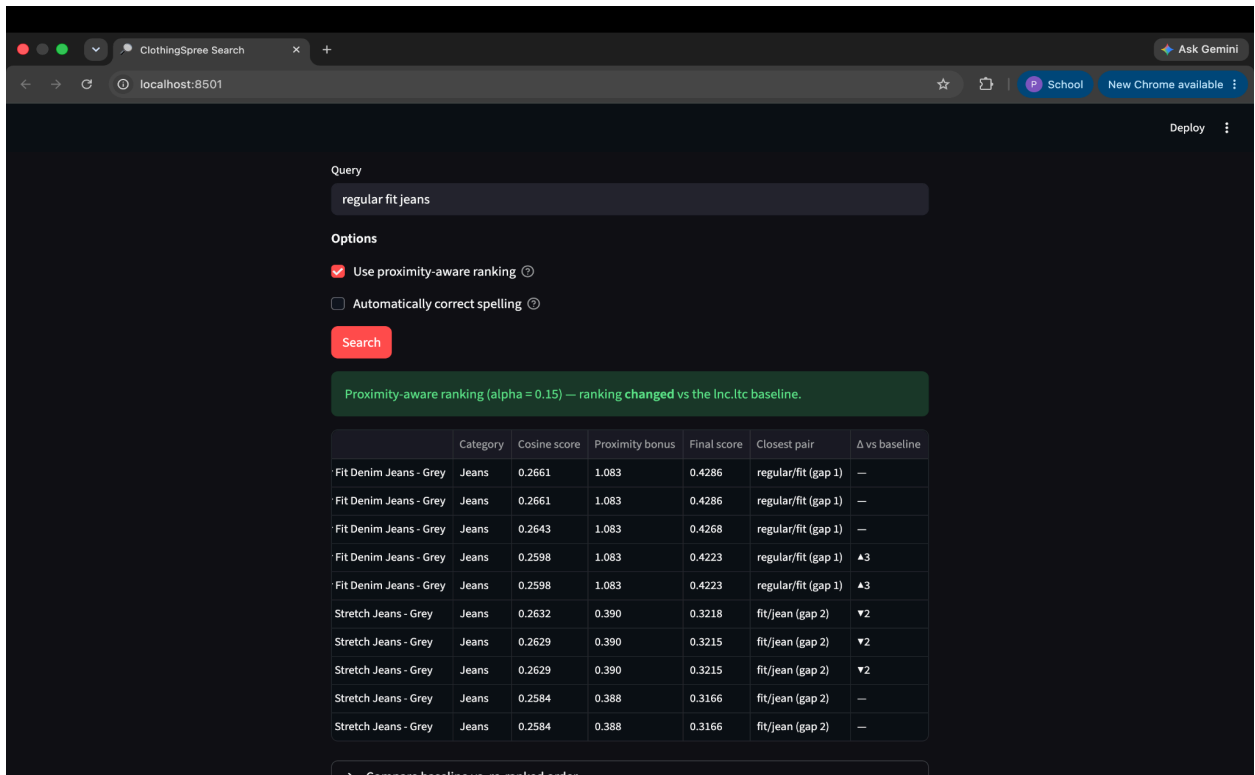
☒ Ordered (term 1 before term 2)

☐ Unordered (either order)

Search proximity

No documents satisfy this constraint. A term may be unknown to the corpus, or the words never occur within the specified distance. Try a larger k or the unordered option.

SECTION 4 — Proximity-Aware Reranking (Positional Info Changes Order)



Query: regular fit jeans

Options:

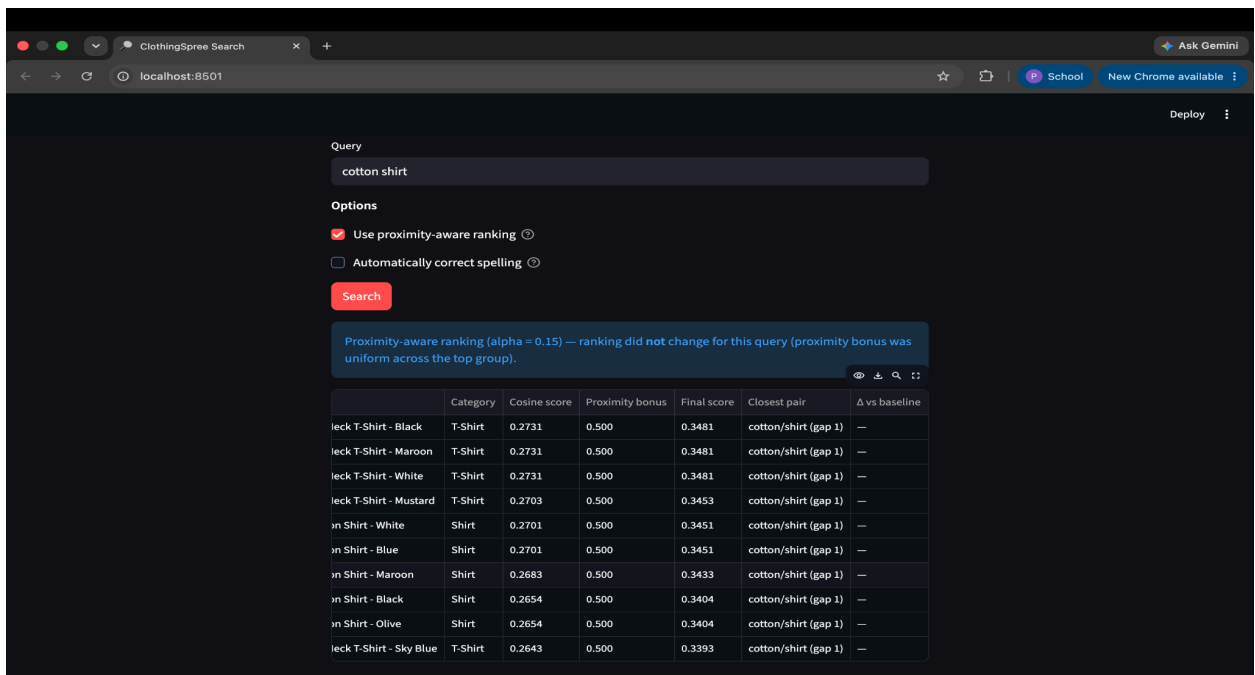
- ☒ Use proximity-aware ranking ⓘ
- ☐ Automatically correct spelling ⓘ

Search

Proximity-aware ranking (alpha = 0.15) — ranking changed vs the Inc.ltc baseline.

	Category	Cosine score	Proximity bonus	Final score	Closest pair	Δ vs baseline
Fit Denim Jeans - Grey	Jeans	0.2661	1.083	0.4286	regular/fit (gap 1)	—
Fit Denim Jeans - Grey	Jeans	0.2661	1.083	0.4286	regular/fit (gap 1)	—
Fit Denim Jeans - Grey	Jeans	0.2643	1.083	0.4268	regular/fit (gap 1)	—
Fit Denim Jeans - Grey	Jeans	0.2598	1.083	0.4223	regular/fit (gap 1)	▲3
Fit Denim Jeans - Grey	Jeans	0.2598	1.083	0.4223	regular/fit (gap 1)	▲3
Stretch Jeans - Grey	Jeans	0.2632	0.390	0.3218	fit/jean (gap 2)	▼2
Stretch Jeans - Grey	Jeans	0.2629	0.390	0.3215	fit/jean (gap 2)	▼2
Stretch Jeans - Grey	Jeans	0.2629	0.390	0.3215	fit/jean (gap 2)	▼2
Stretch Jeans - Grey	Jeans	0.2584	0.388	0.3166	fit/jean (gap 2)	—
Stretch Jeans - Grey	Jeans	0.2584	0.388	0.3166	fit/jean (gap 2)	—

Compare baseline vs re-ranked order



Query: cotton shirt

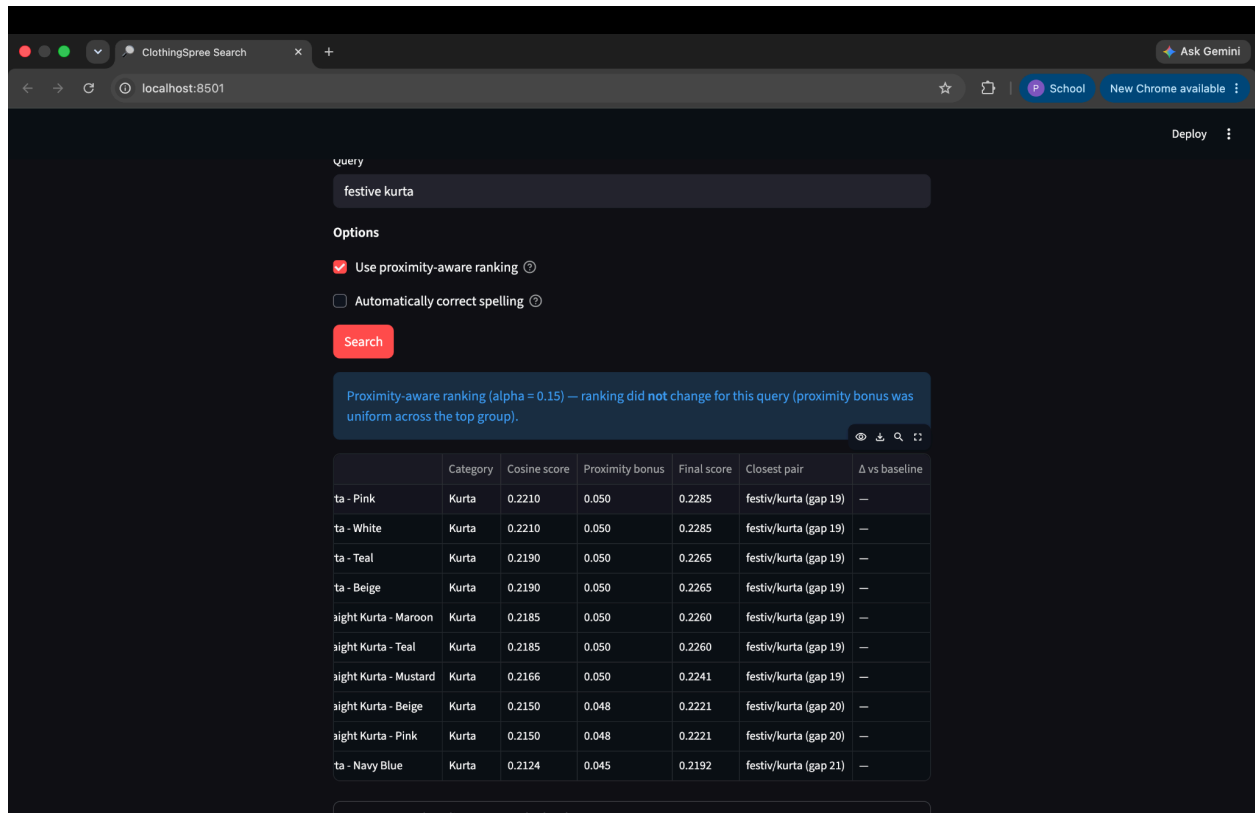
Options:

- ☒ Use proximity-aware ranking ⓘ
- ☐ Automatically correct spelling ⓘ

Search

Proximity-aware ranking (alpha = 0.15) — ranking did not change for this query (proximity bonus was uniform across the top group).

	Category	Cosine score	Proximity bonus	Final score	Closest pair	Δ vs baseline
Jeck T-Shirt - Black	T-Shirt	0.2731	0.500	0.3481	cotton/shirt (gap 1)	—
Jeck T-Shirt - Maroon	T-Shirt	0.2731	0.500	0.3481	cotton/shirt (gap 1)	—
Jeck T-Shirt - White	T-Shirt	0.2731	0.500	0.3481	cotton/shirt (gap 1)	—
Jeck T-Shirt - Mustard	T-Shirt	0.2703	0.500	0.3453	cotton/shirt (gap 1)	—
in Shirt - White	Shirt	0.2701	0.500	0.3451	cotton/shirt (gap 1)	—
in Shirt - Blue	Shirt	0.2701	0.500	0.3451	cotton/shirt (gap 1)	—
in Shirt - Maroon	Shirt	0.2683	0.500	0.3433	cotton/shirt (gap 1)	—
in Shirt - Black	Shirt	0.2654	0.500	0.3404	cotton/shirt (gap 1)	—
in Shirt - Olive	Shirt	0.2654	0.500	0.3404	cotton/shirt (gap 1)	—
Jeck T-Shirt - Sky Blue	T-Shirt	0.2643	0.500	0.3393	cotton/shirt (gap 1)	—



2 Cases where Positional info changes result set/order :

Case 1: Positional Index Narrows the Result Set

```
Docs with BOTH cotton+shirt (co-occurrence): 15
Docs with EXACT phrase cotton shirt:          10
Excluded by phrase search:                    ['D011', 'D031', 'D051', 'D071', 'D091']

D011: Men's Oversized Graphic T-Shirt - Mustard
      cotton @ [11], shirt @ [3, 8, 13] -> gap=2
D031: Men's Oversized Graphic T-Shirt - Olive Green
      cotton @ [13], shirt @ [3, 9, 15] -> gap=2
D051: Men's Oversized Graphic T-Shirt - Navy Blue
      cotton @ [13], shirt @ [3, 9, 15] -> gap=2
D071: Men's Oversized Graphic T-Shirt - Black
      cotton @ [11], shirt @ [3, 8, 13] -> gap=2
D091: Men's Oversized Graphic T-Shirt - Sky Blue
      cotton @ [13], shirt @ [3, 9, 15] -> gap=2
```

The query cotton shirt was run in two modes:

- **Free-text (VSM):** Returns all documents containing "cotton" **or** "shirt" — 63 documents score > 0 (top-10 displayed).
- **Exact phrase search:** Uses the positional index to find documents where "cotton" appears **immediately before** "shirt" (gap = 1). Only **10 documents** match.

15 documents contain both words somewhere, but only **10** have them adjacent. The **5 excluded documents** — D011, D031, D051, D071, D091 — are all "Men's Oversized Graphic T-Shirt" variants. In these documents, "cotton" and "shirt" are separated by a gap of **2 positions** (one token between them), so they never appear consecutively.

For example, in **D011**: cotton is at position 11 and the nearest shirt is at position 13 — gap = 2. A phrase requires gap = 1. The positional index correctly **excludes** D011, while a plain VSM or boolean search would wrongly **include** it.

This demonstrates that positional information **changes the result set** — co-occurrence and exact phrase matching are not equivalent.

Case 2: Positional Index Changes the Ranking Order

```
QUERY: cotton denim

BASELINE order (lnc.ltc cosine only):
Rank 1: D023
Rank 2: D083
Rank 3: D043
Rank 4: D003
Rank 5: D063
Rank 6: D053
Rank 7: D058
Rank 8: D018
Rank 9: D078
Rank 10: D038

RERANKED order (cosine + proximity bonus, alpha=0.15):
Rank 1: D023 cosine=0.2477 proximity_bonus=0.5000 final=0.3227
Rank 2: D083 cosine=0.2477 proximity_bonus=0.5000 final=0.3227
Rank 3: D053 cosine=0.1787 proximity_bonus=0.5000 final=0.2537
Rank 4: D043 cosine=0.2060 proximity_bonus=0.0000 final=0.2060
Rank 5: D003 cosine=0.2024 proximity_bonus=0.0000 final=0.2024
Rank 6: D063 cosine=0.2024 proximity_bonus=0.0000 final=0.2024
Rank 7: D058 cosine=0.1459 proximity_bonus=0.0000 final=0.1459
Rank 8: D018 cosine=0.1449 proximity_bonus=0.0000 final=0.1449
Rank 9: D078 cosine=0.1449 proximity_bonus=0.0000 final=0.1449
Rank 10: D038 cosine=0.1433 proximity_bonus=0.0000 final=0.1433

RANK CHANGES:
D053: ROSE from rank 6 -> rank 3 (cosine=0.1787, bonus=0.5000)
D043: FELL from rank 3 -> rank 4 (cosine=0.2060, bonus=0.0000)
D003: FELL from rank 4 -> rank 5 (cosine=0.2024, bonus=0.0000)
D063: FELL from rank 5 -> rank 6 (cosine=0.2024, bonus=0.0000)
```

Rank	Doc ID	Category	Cosine score	Proximity bonus	Final score	Closest pair	Δ vs baseline
1	D023	Jeans	0.2477	0.500	0.3227	cotton/denim (	—
2	D083	Jeans	0.2477	0.500	0.3227	cotton/denim (	—
3	D053	Jeans	0.1787	0.500	0.2537	cotton/denim (	▲3
4	D043	Jeans	0.2060	0.000	0.2060	—	▼1
5	D003	Jeans	0.2024	0.000	0.2024	—	▼1
6	D063	Jeans	0.2024	0.000	0.2024	—	▼1
7	D058	Jacket	0.1459	0.000	0.1459	—	—
8	D018	Jacket	0.1449	0.000	0.1449	—	—
9	D078	Jacket	0.1449	0.000	0.1449	—	—
10	D038	Jacket	0.1433	0.000	0.1433	—	—

For query cotton denim, the baseline Inc.Itc VSM ranks documents purely by cosine similarity. After enabling proximity-aware reranking ($\alpha = 0.15$), the ranking order changes because the positional index is used to compute how close the query terms appear in each document.

Key change:

- D053 (Men's Slim Fit Stretch Jeans) — cotton and denim appear with gap = 1 (adjacent) → proximity bonus = 0.5000 → final score = $0.1787 + (0.15 \times 0.5000) = 0.2537$ → ROSE from rank 6 → rank 3
- D043 (Men's Regular Fit Denim Jeans) — cotton and denim are far apart → proximity bonus = 0.0000 → final score = 0.2060 (unchanged) → FELL from rank 3 → rank 4

D053 has a lower cosine score (0.1787) than D043 (0.2060), yet it overtakes D043 because positional information reveals that the query terms co-occur closely in D053. This demonstrates that the positional index changes the ranking order — documents where query terms appear close together are considered more relevant, even if their raw term-weight score is lower.

Section 5 - Index Output Screenshots

5a-Dictionary inverted index

```
"100":
  df      : 5
  postings : {'D001': 1, 'D021': 1, 'D041': 1, 'D061': 1, 'D081': 1}

"activ":
  df      : 5
  postings : {'D019': 2, 'D039': 2, 'D059': 2, 'D079': 2, 'D099': 2}

"avail":
  df      : 100
  postings : {'D001': 1, 'D002': 1, 'D003': 1, 'D004': 1, 'D005': 1}

"beig":
  df      : 6
  postings : {'D014': 3, 'D018': 3, 'D048': 3, 'D052': 3, 'D078': 3}

"black":
  df      : 19
  postings : {'D001': 3, 'D002': 3, 'D007': 3, 'D010': 3, 'D019': 3}

"blend":
  df      : 30
  postings : {'D002': 1, 'D004': 1, 'D006': 1, 'D010': 1, 'D011': 1}
```

5b-Postional Index

```
"cotton":  
  df: 58  
  postings:  
    D001: tf=3, positions=[1, 7, 14]  
    D002: tf=2, positions=[2, 7]  
    D005: tf=1, positions=[13]  
    D006: tf=1, positions=[11]  
  
"denim":  
  df: 15  
  postings:  
    D003: tf=3, positions=[3, 9, 15]  
    D013: tf=1, positions=[14]  
    D018: tf=1, positions=[11]  
    D023: tf=3, positions=[3, 9, 14]  
  
"shirt":  
  df: 20  
  postings:  
    D001: tf=3, positions=[4, 10, 15]  
    D002: tf=3, positions=[3, 8, 13]  
    D011: tf=3, positions=[3, 8, 13]  
    D012: tf=3, positions=[3, 8, 13]
```